

Evaluation de performance REOC

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1 Analytic Results

Question 1: Parameters

We set $N = 1$, $\lambda_H = 2$, $\mu_H = \mu_R = \mu_D = 1$, $C = 1$.

Question 1(a)(i): Analytical calculation with $p = 1$

With $p = 1$, node D is never visited. The network is reduced to three nodes: Thread (T), HTTP (H), and Resources (R), each modeled as a single-server queue.

Solution of the traffic equations

Let's set $\gamma_R = 1$. The traffic equations yield:

$$\gamma_T = \gamma_H = p \gamma_R = 1, \quad \gamma_D = (1 - p) \gamma_R = 0.$$

The loads are:

$$\rho_T = \frac{\gamma_T}{\lambda_H} = \frac{1}{2}, \quad \rho_H = \frac{\gamma_H}{\mu_H} = 1, \quad \rho_R = \frac{\gamma_R}{\mu_R} = 1.$$

Stationary distribution

With $N = 1$, the possible states are $(n_T, n_H, n_R) \in \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$. The joint distribution gives:

$$\pi(n_T, n_H, n_R) = \frac{1}{G} \rho_T^{n_T} \rho_H^{n_H} \rho_R^{n_R},$$

with the normalization constant:

$$G = \rho_T + \rho_H + \rho_R = \frac{1}{2} + 1 + 1 = \frac{5}{2}.$$

We get:

$$\pi(1, 0, 0) = \frac{1/2}{5/2} = \frac{1}{5}, \quad \pi(0, 1, 0) = \frac{1}{5/2} = \frac{2}{5}, \quad \pi(0, 0, 1) = \frac{1}{5/2} = \frac{2}{5}.$$

Probability of rejection

A request is rejected if $n_T = 0$:

$$P_{\text{rej}} = \pi(0, 1, 0) + \pi(0, 0, 1) = \frac{2}{5} + \frac{2}{5} = \frac{4}{5} = 0.8$$

Throughput

$$\theta_H = \theta_R = \theta_T = \pi_T(1) \cdot \lambda_H = \frac{1}{5} \times 2 = \frac{2}{5}.$$

Mean number of requests per station

$$\bar{A}_T = \pi_T(1) = \frac{1}{5}, \quad \bar{A}_H = \pi_H(1) = \frac{2}{5}, \quad \bar{A}_R = \pi_R(1) = \frac{2}{5}.$$

Mean sojourn time

By Little's law, $\bar{B}_i = \bar{A}_i / \theta_i$:

$$\bar{B}_H = \frac{2/5}{2/5} = 1, \quad \bar{B}_R = \frac{2/5}{2/5} = 1.$$

With $p = 1$, the $R + D$ subsystem reduces to R alone:

$$\bar{B}_{R+D} = \frac{\bar{B}_R + (1-p)\bar{B}_D}{p} = \frac{1+0}{1} = 1.$$

The total mean sojourn time of served requests is:

$$\boxed{\bar{B} = \bar{B}_H + \bar{B}_{R+D} = 1 + 1 = 2.}$$

Question 1(b): Doubling the capacity of a server

We consider the general case $p \in (0, 1]$, $N = 1$, $\lambda_H = 2$, $\mu_H = \mu_R = \mu_D = 1$, $C = 1$.

Setup: general case

The normalized traffic solution ($\gamma_R = 1$) gives:

$$\gamma_T = \gamma_H = p, \quad \gamma_D = 1 - p,$$

and the loads are:

$$\rho_T = \frac{p}{\lambda_H} = \frac{p}{2}, \quad \rho_H = p, \quad \rho_R = 1, \quad \rho_D = 1 - p.$$

The normalization constant with $N = 1$ is:

$$G = \rho_T + \rho_H + \rho_R + \rho_D = \frac{p}{2} + p + 1 + (1 - p) = 2 + \frac{p}{2}.$$

The probability that node T is busy (i.e., a request is being processed):

$$\pi_T(1) = \frac{\rho_T}{G} = \frac{p/2}{2 + p/2} = \frac{p}{4 + p}.$$

Probability of rejection (reference):

$$P_{\text{rej}}^{(0)} = 1 - \pi_T(1) = 1 - \frac{p}{4 + p} = \frac{4}{4 + p}.$$

Mean sojourn time (reference):

The throughput is $\theta = \pi_T(1) \cdot \lambda_H = \frac{p}{4+p} \cdot 2 = \frac{2p}{4+p}$. By Little's law, $\bar{B}_i = \pi_i(1) / \theta_i$ where $\theta_i = \pi_i(1) \cdot \mu_i$, thus $\bar{B}_i = 1 / \mu_i$ for $i \in \{H, R, D\}$. Thus:

$$\bar{B}^{(0)} = \frac{1}{\mu_H} + \frac{1/\mu_R + (1-p)/\mu_D}{p} = 1 + \frac{1 + (1-p)}{p} = 1 + \frac{2-p}{p} = \frac{p+2-p}{p} + 1 = \frac{2}{p}.$$

Remark: the mean sojourn time $\bar{B}^{(0)} = 2/p$ does not depend on the μ_i 's but only on p , because with $N = 1$ each server is either empty or busy serving the single request.

Calculation of the four doubling options

We successively double λ_H (server T), μ_H , μ_R , or μ_D . In each case, we recalculate G' , P'_{rej} , and $\bar{B}'$.

Option A: double λ_H (server T). $\rho'_T = p/(2\lambda_H) = p/4$.

$$G' = \frac{p}{4} + p + 1 + (1 - p) = 2 + \frac{p}{4}.$$

$$P_{\text{rej}}^A = 1 - \frac{p/4}{2 + p/4} = \frac{2}{2 + p/4} \cdot \frac{1}{1} = \frac{8}{8 + p}.$$

$$\bar{B}^A = \frac{1}{\mu_H} + \frac{1/\mu_R + (1 - p)/\mu_D}{p} = \frac{2}{p} \quad (\text{unchanged because the } \mu_i \text{'s do not change}).$$

Option B: double μ_H . $\rho'_H = p/(2\mu_H) = p/2$.

$$G' = \frac{p}{2} + \frac{p}{2} + 1 + (1 - p) = 2.$$

$$P_{\text{rej}}^B = 1 - \frac{p/2}{2} = 1 - \frac{p}{4} = \frac{4 - p}{4}.$$

$$\bar{B}^B = \frac{1}{2\mu_H} + \frac{1/\mu_R + (1 - p)/\mu_D}{p} = \frac{1}{2} + \frac{2 - p}{p}.$$

Option C: double μ_R . $\rho'_R = 1/(2\mu_R) = 1/2$.

$$G' = \frac{p}{2} + p + \frac{1}{2} + (1 - p) = \frac{3}{2} + \frac{p}{2}.$$

$$P_{\text{rej}}^C = 1 - \frac{p/2}{3/2 + p/2} = \frac{3/2}{3/2 + p/2} = \frac{3}{3 + p}.$$

$$\bar{B}^C = \frac{1}{\mu_H} + \frac{1/(2\mu_R) + (1 - p)/\mu_D}{p} = 1 + \frac{1/2 + (1 - p)}{p} = 1 + \frac{3/2 - p}{p} = \frac{3}{2p}.$$

Option D: double μ_D . $\rho'_D = (1 - p)/(2\mu_D) = (1 - p)/2$.

$$G' = \frac{p}{2} + p + 1 + \frac{1 - p}{2} = \frac{3}{2} + \frac{p}{2} + \frac{1}{2} = 2 + \frac{p - 1}{2}.$$

Let us verify: $G' = p/2 + p + 1 + (1 - p)/2 = p/2 + p + 1 + 1/2 - p/2 = p + 3/2$.

$$P_{\text{rej}}^D = 1 - \frac{p/2}{p + 3/2} = \frac{p + 3/2 - p/2}{p + 3/2} = \frac{p/2 + 3/2}{p + 3/2} = \frac{p + 3}{2p + 3}.$$

$$\bar{B}^D = 1 + \frac{1/\mu_R + (1 - p)/(2\mu_D)}{p} = 1 + \frac{1 + (1 - p)/2}{p} = 1 + \frac{(3 - p)/2}{p} = 1 + \frac{3 - p}{2p} = \frac{2p + 3 - p}{2p} = \frac{p + 3}{2p}.$$

(i) Reduce the probability of rejection

We compare the four probabilities of rejection:

Option	Doubled server	P'_{rej}
Reference (initial)	—	$\frac{4}{4 + p}$
A	λ_H (T)	$\frac{8}{8 + p}$
B	μ_H (H)	$\frac{4 - p}{4}$
C	μ_R (R)	$\frac{3}{3 + p}$
D	μ_D (D)	$\frac{p + 3}{2p + 3}$

The probability of rejection is $P_{\text{rej}} = 4/(4+p)$ in the reference case. To reduce P_{rej} , we look for the option that gives the smallest value.

Option A vs reference: $\frac{8}{8+p} < \frac{4}{4+p} \Leftrightarrow 8(4+p) < 4(8+p) \Leftrightarrow 32+8p < 32+4p \Leftrightarrow 4p < 0$, false.

In reality: $8/(8+p) = 4/(4+p/2) < 4/(4+p)$ since $p/2 < p$. Therefore, option A always improves P_{rej} .

Comparison A vs C:

$$\frac{8}{8+p} \text{ vs } \frac{3}{3+p}.$$

$$\frac{8}{8+p} < \frac{3}{3+p} \Leftrightarrow 8(3+p) < 3(8+p) \Leftrightarrow 24+8p < 24+3p \Leftrightarrow 5p < 0, \text{ false.}$$

Thus $\frac{3}{3+p} < \frac{8}{8+p}$ for all $p > 0$: **option C (doubling μ_R) yields a lower probability of rejection than option A.**

Comparison C vs B:

$$\frac{3}{3+p} \text{ vs } \frac{4-p}{4}.$$

$$\frac{3}{3+p} < \frac{4-p}{4} \Leftrightarrow 12 < (4-p)(3+p) = 12+p-p^2 \Leftrightarrow p-p^2 > 0 \Leftrightarrow p(1-p) > 0, \text{ which is true for } p \in (0, 1).$$

For $p = 1$: $3/4$ vs $3/4$, equality.

Thus **option C is better than B** for $p \in (0, 1)$, and equivalent for $p = 1$.

Comparison C vs D:

$$\frac{3}{3+p} \text{ vs } \frac{p+3}{2p+3}.$$

$$\frac{3}{3+p} < \frac{p+3}{2p+3} \Leftrightarrow 3(2p+3) < (p+3)^2 \Leftrightarrow 6p+9 < p^2+6p+9 \Leftrightarrow 0 < p^2, \text{ true for all } p \in (0, 1].$$

Thus **option C is better than D** for all $p > 0$.

Conclusion (i): For all $p \in (0, 1]$, doubling the capacity of server R (option C) minimizes the probability of rejection.

(ii) Reduce the mean sojourn time

Option	Doubled server	$\bar{B}'$
Reference	–	$\frac{2}{p}$
A	λ_H (T)	$\frac{2}{p}$ (unchanged)
B	μ_H (H)	$\frac{1}{2} + \frac{2-p}{p}$
C	μ_R (R)	$\frac{3}{2p}$
D	μ_D (D)	$\frac{p+3}{2p}$

Remark: Option A (doubling λ_H) has no effect on $\bar{B}$ because the mean sojourn time depends only on the service rates μ_i .

Comparison C vs B:

$$\frac{3}{2p} \text{ vs } \frac{1}{2} + \frac{2-p}{p} = \frac{p+2(2-p)}{2p} = \frac{4-p}{2p}.$$

$$\frac{3}{2p} < \frac{4-p}{2p} \Leftrightarrow 3 < 4-p \Leftrightarrow p < 1.$$

Thus for $p < 1$, option C is better than B; for $p = 1$, equality ($\bar{B} = 3/2$).

Comparison C vs D:

$$\frac{3}{2p} \text{ vs } \frac{p+3}{2p}.$$

$$\frac{3}{2p} < \frac{p+3}{2p} \Leftrightarrow 3 < p+3 \Leftrightarrow p > 0.$$

Thus option C is **always strictly better** than D.

Case $p = 1$:

$$\bar{B}^B = \frac{1}{2} + 1 = \frac{3}{2}, \quad \bar{B}^C = \frac{3}{2}, \quad \bar{B}^D = 2.$$

Options B and C are equivalent and the best.

Conclusion (ii):

- For $p \in (0, 1)$: doubling μ_R (**option C**) minimizes the mean sojourn time.
- For $p = 1$: doubling μ_H or μ_R are equivalent (both yield $\bar{B} = 3/2$), and better than the other options.

2 Simulation

The following results are issued with the simulator.py code (attached file)

2.1 Question 1 (a) (ii)

For the given parameters ($N = 1$, $\lambda_H = 2$, $\mu_i = 1$, and $p = 1$), the simulation yielded the following metrics:

- **Measured probability of rejection:** ≈ 0.7979
- **Measured mean sojourn time:** ≈ 1.9858 s

These empirical results strongly corroborate the analytical expressions derived in Question 1(a)(i), where the theoretical values were exactly $P_{\text{rej}} = 0.8$ (or $4/5$) and $\bar{B} = 2$ s.

2.2 Question 2

Mean Sojourn Time ($\bar{B}$)	(a) $N = 15, p = 0.5$	(b) $N = 15, p = 0.22$	(c) $N = 30, p = 0.5$	(d) $N = 30, p = 0.22$
Closed Network	2.266 s	7.844 s	2.252 s	11.414 s
Open Network	2.246 s	14.251 s	2.266 s	13.064 s
Approx 1 (Inf. threads)	2.262 s	13.035 s	2.262 s	13.035 s
Approx 2 (Single res.)	0.889 s	10.660 s	0.889 s	10.660 s
Approx 3 (Inf. res.)	2.222 s	5.206 s	2.222 s	5.206 s
Reject. Prob. (Closed)	0.002 %	3.996 %	0.000 %	0.809 %

Final Conclusion

- **Impact of N :** A small thread pool ($N = 15$) protects the network from congestion by rejecting excess traffic at the entry. A large pool ($N = 30$) accepts almost everything, which shifts the delays inside the network. **As N grows, the closed network behaves exactly like an open network.**
- **Approximation 1 (Infinite threads):** Highly accurate for a large N , but fails for a small N because it completely ignores request rejections.
- **Approximation 2 (Bottleneck):** Only valid under **heavy load** ($p = 0.22$) because it successfully models the massive congestion at the Resource server.
- **Approximation 3 (Infinite resources):** Only valid under **light load** ($p = 0.5$). It completely fails under heavy load because it naively assumes physical queues never form.

3 Notice

An artificial intelligence model (Google Gemini) was used to assist in the preparation of this report, specifically for translating into English, formatting the document in LaTeX, and cleaning up the Python source code.